

# ECON 0100 | Vignette B5 | Market Changes

*Due in Recitation. Vignettes are a certificate of the work done together in Recitation.*

Members of the wizarding world have preferences for pumpkin pasties according to the following demand curve:

$$P_d = 14 - \left(\frac{1}{2}\right)Q_d$$

Pumpkin pasties are produced by many sellers according to the following supply curve:

$$P = 2 + \left(\frac{1}{2}\right)Q_s$$

Prices are in galleons and quantity is in pasties.

## Q1 | Price Elasticity

- a) Use the midpoint method to find the elasticity of demand when the price changes from 6 to 8 galleons. \_\_\_\_\_
- b) Is demand elastic, unit elastic, or inelastic in a)? \_\_\_\_\_
- c) Use the midpoint method to find the elasticity of demand when the price changes from 1 to 3 galleons. \_\_\_\_\_
- d) Use the midpoint method to find the elasticity of supply when the price changes from 4 to 6 galleons. \_\_\_\_\_

## Q2 | Supply & Demand Shifters

Recent drier growing seasons has made growing pumpkins more difficult, with a new supply curve that can be represented by the following. At a price of 8 galleons, how has the quantity supplied and producer surplus changed with this change in climate?

$$P = 4 + \left(\frac{1}{2}\right)Q_s$$

a) Change in quantity supplied: \_\_\_\_\_

b) Change in producer surplus: \_\_\_\_\_

A popular channel on the wizarding social network FlueTube has been promoting pumpkin pasties, leading to an increase in the popularity of the snack.

c) The demand curve: *shifted in, stayed the same, shifted out*

d) The supply curve: *shifted in, stayed the same, shifted out*

### Q3 | Comparative Statics

Using the new supply curve from Q2 and the original demand curve, find and plot the equilibrium price and quantity for pumpkin pasties.

a) Equilibrium price: \_\_\_\_\_

b) Equilibrium quantity: \_\_\_\_\_

With both the drier growing seasons and the FlueTube promotion, without using numbers, use the graph above to discuss how the market has been impacted.

c) Equilibrium price has: *increased, stayed the same, decreased, indeterminate*

d) Equilibrium quantity has: *increased, stayed the same, decreased, indeterminate*